

=== Section 1: Raw Google Matrix G ===

	A	B	C	D	E	F	G	H	I	J	K	L
A	0.0000	0.5000	0.0000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
B	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
C	0.3333	0.3333	0.0000	0.0000	0.0000	0.0000	0.0000	0.3333	0.0000	0.0000	0.0000	0.0000
D	0.0000	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
E	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.0000	0.0000
F	0.2500	0.0000	0.0000	0.2500	0.0000	0.0000	0.2500	0.0000	0.0000	0.0000	0.2500	0.0000
G	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.0000
H	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000	0.0000	0.0000
I	0.0000	0.0000	0.0000	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.2500	0.2500	0.0000
J	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000
K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000
L	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Row sums: A=1.000000, B=1.000000, C=1.000000, D=1.000000, E=1.000000, F=1.000000, G=1.000000, H=1.000000, I=1.000000, J=1.000000, K=1.000000, L=0.000000												

=== Section 2: Fix Dangling Nodes ===

Dangling pages: L

	A	B	C	D	E	F	G	H	I	J	K	L
A	0.0000	0.5000	0.0000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
B	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
C	0.3333	0.3333	0.0000	0.0000	0.0000	0.0000	0.0000	0.3333	0.0000	0.0000	0.0000	0.0000
D	0.0000	0.0000	0.0000	0.0000	0.5000	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
E	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.0000	0.0000
F	0.2500	0.0000	0.0000	0.2500	0.0000	0.0000	0.2500	0.0000	0.0000	0.0000	0.2500	0.0000
G	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.0000
H	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000	0.0000	0.0000
I	0.0000	0.0000	0.0000	0.0000	0.0000	0.2500	0.2500	0.0000	0.0000	0.2500	0.2500	0.0000
J	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000
K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	0.5000
L	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833
Row sums: A=1.000000, B=1.000000, C=1.000000, D=1.000000, E=1.000000, F=1.000000, G=1.000000, H=1.000000, I=1.000000, J=1.000000, K=1.000000, L=1.000000												

=== Section 3: Eigendecomposition of G^T ===

Eigenvalues of the transposed matrix:

1: 1.0000000000+0.0000000000j
2: -0.0982343045+0.4967099232j
3: -0.0982343045-0.4967099232j
4: -0.4883845314+0.2049905087j
5: -0.4883845314-0.2049905087j
6: 0.4354779012+0.0000000000j
7: 0.2694426423+0.1426693282j
8: 0.2694426423-0.1426693282j
9: -0.3852271966+0.0000000000j
10: -0.2251282038+0.0000000000j
11: 0.0000000000+0.0000000000j
12: -0.1074367803+0.0000000000j

PageRank vector:

A: 0.0468562021
B: 0.0402006515
C: 0.0329048735
D: 0.1649957435
E: 0.1084024568
F: 0.1203346068
G: 0.1169575832
H: 0.0438731646
I: 0.1281299028
J: 0.0597733173
K: 0.0679203867
L: 0.0696511113
L1 norm: 1.0000000000

Ranking: D > I > F > G > E > L > K > J > A > H > B > C

=== Section 4: Power Method for G ===

Iterations: 35

Final successive-iterate L1 difference: 5.985978379641e-11

Converged PageRank vector:

A: 0.0468562021

B: 0.0402006515

C: 0.0329048735

D: 0.1649957435

E: 0.1084024568

F: 0.1203346068

G: 0.1169575832

H: 0.0438731646

I: 0.1281299028

J: 0.0597733173

K: 0.0679203867

L: 0.0696511113

L1 norm: 1.0000000000

L1 difference from eigendecomposition: 1.748036437821e-11

=== Section 5: Teleportation Matrix G_{tilde} ===

$\alpha = 0.85$

	A	B	C	D	E	F	G	H	I	J	K	L
A	0.0125	0.4375	0.0125	0.4375	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125
B	0.0125	0.0125	0.0125	0.4375	0.4375	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125
C	0.2958	0.2958	0.0125	0.0125	0.0125	0.0125	0.0125	0.2958	0.0125	0.0125	0.0125	0.0125
D	0.0125	0.0125	0.0125	0.0125	0.4375	0.4375	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125
E	0.0125	0.0125	0.2250	0.2250	0.0125	0.0125	0.2250	0.2250	0.0125	0.0125	0.0125	0.0125
F	0.2250	0.0125	0.0125	0.2250	0.0125	0.0125	0.2250	0.0125	0.0125	0.0125	0.2250	0.0125
G	0.0125	0.0125	0.0125	0.4375	0.0125	0.0125	0.0125	0.0125	0.4375	0.0125	0.0125	0.0125
H	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.4375	0.0125	0.0125	0.4375	0.0125	0.0125
I	0.0125	0.0125	0.0125	0.0125	0.0125	0.2250	0.2250	0.0125	0.0125	0.2250	0.2250	0.0125
J	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.4375	0.0125	0.0125	0.4375
K	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.4375	0.0125	0.0125	0.4375
L	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833	0.0833

Row sums: A=1.000000, B=1.000000, C=1.000000, D=1.000000, E=1.000000, F=1.000000, G=1.000000, H=1.000000, I=1.000000, J=1.000000, K=1.000000, L=1.000000

=== Section 6: Eigendecomposition of G_tilde^T ===

Eigenvalues of the transposed matrix:

```
1: 1.0000000000+0.0000000000j
2: -0.0834991588+0.4222034348j
3: -0.0834991588-0.4222034348j
4: -0.4151268517+0.1742419324j
5: -0.4151268517-0.1742419324j
6: 0.3701562161+0.0000000000j
7: 0.2290262460+0.1212689290j
8: 0.2290262460-0.1212689290j
9: -0.3274431171+0.0000000000j
10: -0.1913589732+0.0000000000j
11: -0.0000000000+0.0000000000j
12: -0.0913212633+0.0000000000j
```

PageRank vector with teleportation:

```
A: 0.0522131504
B: 0.0512578588
C: 0.0400436037
D: 0.1542085532
E: 0.1050448073
F: 0.1089217905
G: 0.1106915062
H: 0.0513892914
I: 0.1207603443
J: 0.0652236042
K: 0.0665290358
L: 0.0737164542
L1 norm: 1.0000000000
```

Ranking: D > I > G > F > E > L > K > J > A > H > B > C

Position	G page	G score	G_tilde page	G_tilde score
1	D	0.1649957435	D	0.1542085532
2	I	0.1281299028	I	0.1207603443
3	F	0.1203346068	G	0.1106915062
4	G	0.1169575832	F	0.1089217905
5	E	0.1084024568	E	0.1050448073

6	L	0.0696511113	L	0.0737164542
7	K	0.0679203867	K	0.0665290358
8	J	0.0597733173	J	0.0652236042
9	A	0.0468562021	A	0.0522131504
10	H	0.0438731646	H	0.0513892914
11	B	0.0402006515	B	0.0512578588
12	C	0.0329048735	C	0.0400436037

=== Section 7: Power Method for G_{tilde} ===

Iterations: 28

Final successive-iterate L1 difference: 5.959566867775e-11

Converged PageRank vector:

A: 0.0522131504

B: 0.0512578588

C: 0.0400436037

D: 0.1542085532

E: 0.1050448073

F: 0.1089217905

G: 0.1106915062

H: 0.0513892914

I: 0.1207603443

J: 0.0652236042

K: 0.0665290358

L: 0.0737164542

L1 norm: 1.0000000000

L1 difference from eigendecomposition: 1.700200397137e-11

=== Section 8: Term-Document Matrix T ===

Dimensions: 15 terms x 12 documents

	A	B	C	D	E	F	G	H	I	J	K	L
Alfa Romeo	0	0	0	1	0	1	1	1	0	1	1	0
Audi	1	0	0	0	0	0	1	1	0	1	0	1
Cadillac	0	1	0	0	1	0	0	1	1	1	1	0
Dodge	1	1	0	0	1	0	0	0	1	0	0	0
Hyundai	1	1	0	1	0	0	0	1	1	0	1	0
Jaguar	1	1	1	0	1	0	1	0	1	0	1	1
Lexus	1	0	0	1	0	0	1	1	0	1	0	1
Mazda	1	0	1	0	0	1	1	0	0	1	1	1
Mercedes	1	1	1	1	0	1	0	0	0	0	0	1
Nissan	1	0	1	0	0	1	0	0	1	0	0	1
Porsche	1	0	1	1	1	1	1	0	0	1	1	0
Subaru	0	1	0	1	1	0	0	0	0	1	1	1
Tesla	0	1	0	1	0	1	1	0	0	0	1	1
Toyota	0	1	1	0	1	0	0	1	1	1	0	1
Volkswagen	0	0	1	0	1	1	0	1	1	0	0	0

=== Section 9: Search Queries ===

Query (SINGLE): Nissan

q for Nissan (terms in the Section 8 order): 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0

d for Nissan: A=1, B=0, C=1, D=0, E=0, F=1, G=0, H=0, I=1, J=0, K=0, L=1

Matching pages: A, C, F, I, L

Ranking by G_tilde PageRank: I > F > L > A > C

1. I: 0.1207603443
2. F: 0.1089217905
3. L: 0.0737164542
4. A: 0.0522131504
5. C: 0.0400436037

Query (AND): Mercedes AND Jaguar

q for Mercedes (terms in the Section 8 order): 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0

d for Mercedes: A=1, B=1, C=1, D=1, E=0, F=1, G=0, H=0, I=0, J=0, K=0, L=1

q for Jaguar (terms in the Section 8 order): 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0

d for Jaguar: A=1, B=1, C=1, D=0, E=1, F=0, G=1, H=0, I=1, J=0, K=1, L=1

Matching pages: A, B, C, L

Ranking by G_tilde PageRank: L > A > B > C

1. L: 0.0737164542
2. A: 0.0522131504
3. B: 0.0512578588
4. C: 0.0400436037

Query (OR): Toyota OR Porsche

q for Toyota (terms in the Section 8 order): 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0

d for Toyota: A=0, B=1, C=1, D=0, E=1, F=0, G=0, H=1, I=1, J=1, K=0, L=1

q for Porsche (terms in the Section 8 order): 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0

d for Porsche: A=1, B=0, C=1, D=1, E=1, F=1, G=1, H=0, I=0, J=1, K=1, L=0

Matching pages: A, B, C, D, E, F, G, H, I, J, K, L

Ranking by G_tilde PageRank: D > I > G > F > E > L > K > J > A > H > B > C

1. D: 0.1542085532
2. I: 0.1207603443
3. G: 0.1106915062
4. F: 0.1089217905

5. E: 0.1050448073
6. L: 0.0737164542
7. K: 0.0665290358
8. J: 0.0652236042
9. A: 0.0522131504
10. H: 0.0513892914
11. B: 0.0512578588
12. C: 0.0400436037

Query (BUT NOT): Mazda BUT NOT Volkswagen

q for Mazda (terms in the Section 8 order): 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0

d for Mazda: A=1, B=0, C=1, D=0, E=0, F=1, G=1, H=0, I=0, J=1, K=1, L=1

q for Volkswagen (terms in the Section 8 order): 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1

d for Volkswagen: A=0, B=0, C=1, D=0, E=1, F=1, G=0, H=1, I=1, J=0, K=0, L=0

Matching pages: A, G, J, K, L

Ranking by G_tilde PageRank: G > L > K > J > A

1. G: 0.1106915062
2. L: 0.0737164542
3. K: 0.0665290358
4. J: 0.0652236042
5. A: 0.0522131504